



Service Bulletin

Service Bulletin Number: 6628629

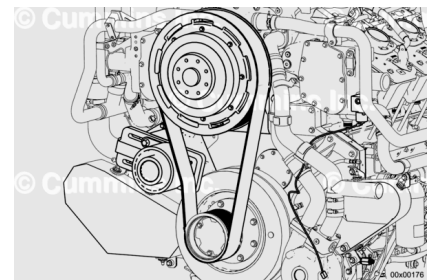
Released Date: 06-may-2025

Rockford® Fan Clutch

Rockford® Fan Clutch

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Example Installation: QSK78 with Rockford clutch installed.



Product Affected

- KTA50
- KTTA50
- QSK38 CM2150 MCRS
- QSK38 CM2350 K125
- QSK38 CM850 MCRS
- QSK45 CM500
- QSK50 CM2150 MCRS
- QSK50 CM2350 K108
- QSK50 CM2350 K130
- QSK50 CM850 MCRS
- QSK60 CM2150 MCRS
- QSK60 CM2350 K116
- QSK60 CM2350 K121
- QSK60 CM2350 K135
- QSK60 CM2350 K136
- QSK60 CM500
- QSK60 CM850 MCRS
- QSK78 CM2350 K126
- QSK78 CM2350 K137
- QSK78 CM2350 K154C
- QSK78 CM500

Overhaul Components

Table 1, Overhaul Components

Kit Part Number	Rockford® Clutch Cummins® Part Number(s):	Engine Service Model(s):
Overhaul kit with Clutch Plates (5540992)	4016984 4016983	QSK60 QSK78
Seal Kit (5540993)	4016984 4016983 3626879	QSK60 QSK78 K/QSK38 K/QSK50
	3627923	QSK50
Overhaul Kit without Clutch Plates (5540996)	3626879	K/QSK38 K/QSK50
	3627923	QSK50
Overhaul Kit with Clutch Plates (5540997)	3626879	K/QSK38 K/QSK50
	3627923	QSK50

Warranty

- The warranty covering the equipment will be invalid in the event of damage caused by unspecified working practices and/or failure to observe the correct specified procedures.
- Any relevant classification society or local authority regulations **must** be followed.

General Description

- The Rockford® Fan Drive Clutch is an oil pressure actuated, oil cooled and lubricated, multi-plate clutch designed for infinite fan-to-engine pulley speed ratios, assuring prescribed engine coolant temperatures and minimum engine horsepower losses. Engine oil is supplied to the fan clutch to provide maximum cooling and lubrication. Engine cooling temperature demands are transmitted to the clutch either by a Rockford® fan clutch thermal sensor located to sense the engine coolant temperature, or by the engine's electronic control system. Optional sensors may be used to control temperatures in other circuits. The fan clutch automatically adjusts the fan speed to maintain specified coolant temperature. It is controlled by the thermal sensors and solenoid valve which cause the fan speed to be increased or decreased smoothly without shock loads. Oil cooled plates permit clutch slippage which allows variable fan speeds.
- The fan clutch shaft acts as a bearing surface for the moving parts. Internal ports and orifices distribute engine oil to lubricate, cool and engage the clutch plates which control fan speed.

Overhaul Tools

- Rockford Clutch® Service Tools are **not** available for sale.

- Rockford Clutch® Service Tools are to be made by a competent engineering shop to the specification outlined in the technical drawings in the “Overhaul Tool Drawings” section at the end of this document.

Component Parts (Exploded View)

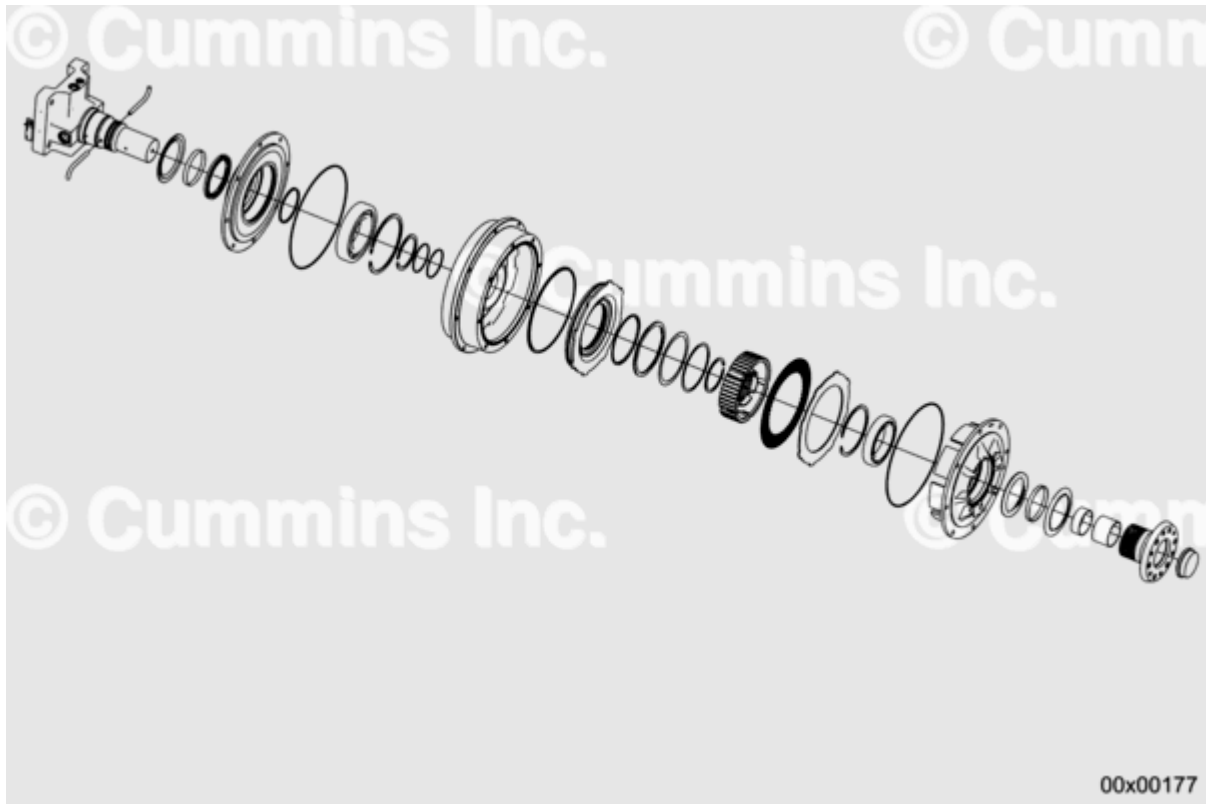


Figure 1, Representative Exploded View Diagram. Similar to Part Number 3626879.

Overhaul

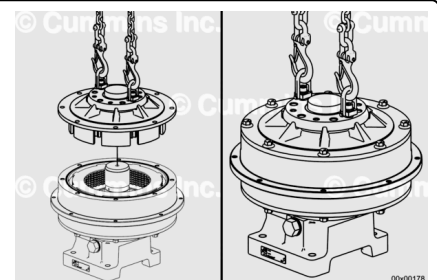
Disassemble

- Follow the appropriate preparatory steps to remove the Rockford® Clutch from the engine before beginning the overhaul procedure.
- See corresponding Service Manual. Reference Procedure 008-036 in Section 8.

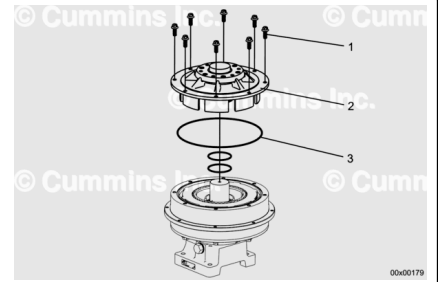


- If the pulley is a two-piece bolt-on design, remove the outer ring of bolts and washers. Separate the pulley from the pulley adapter to reduce weight.
- Support beneath the pulley and adapter when removing capscrews.

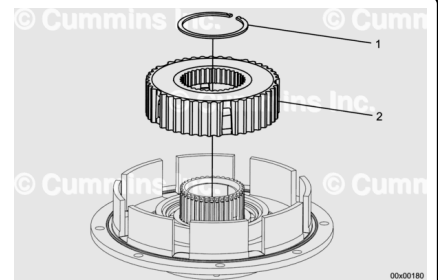
Note : The pulley may be tight on the pulley adapter, gently tap with a hammer if required.



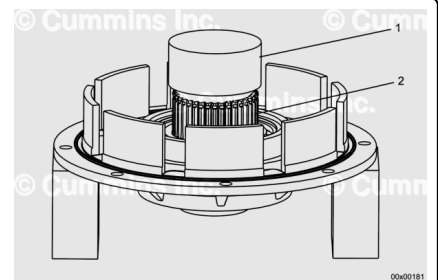
- Position the Rockford® Clutch on the work area with the fan mounting hub up. Support beneath the pulley/adaptor.
- Remove capscrews and washers (1). Use a small screwdriver at the split-line to break the bearing retainer loose.
- Lift the front bearing retainer (2) off the assembly.
- Remove and discard the "o" ring seal (3).



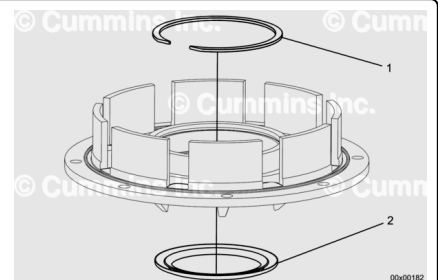
- Rotate the front bearing retainer and position face down on a working surface.
- Remove external snapping (1).
- Lift off the clutch hub (2).



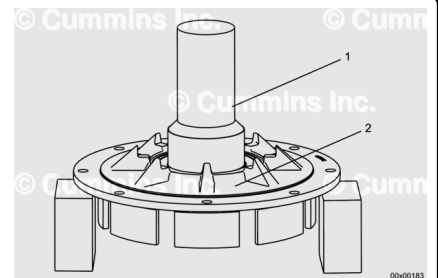
- Position the assembly under a hydraulic press. Support the front bearing retainer assembly to allow the fan mounting hub assembly to be suspended.
- Use tool 1B (1) to remove the fan mounting hub assembly (2).



Remove the internal snapping (1) and front oil seal (2).

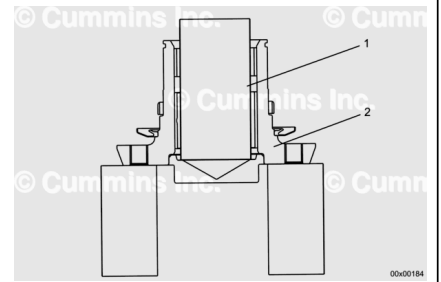


- Position the assembly under a hydraulic press.
- Orient the front bearing retainer (2) so the front face is upwards.
- Support the assembly to avoid pressing on the clutch disc tangs.
- Use tool 2 (1) to remove the front main bearing.

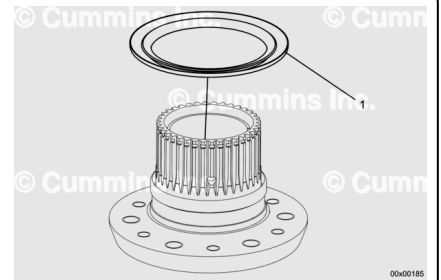


- Remove the end cap, if installed.

- Position the fan mounting hub assembly (2) on the hydraulic press and support beneath the fan mounting flange.
- Press out the end cap using tool 5 (1).



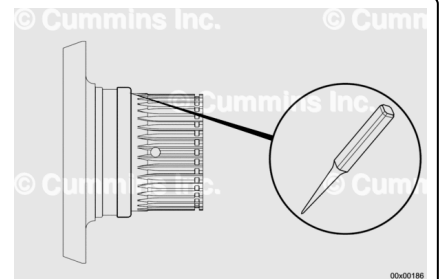
- Remove the seal retainer (1) from the fan mounting hub.
- Use a small pry bar or screwdriver and lever between the rear of the fan mounting flange and the seal retainer to break the press fit free.



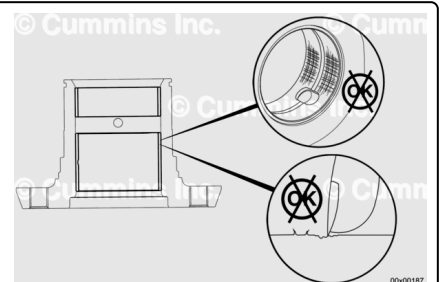
⚠ CAUTION ⚠

Do not cut through the wear sleeve with the chisel. Damage to the fan hub shaft will occur, causing seal leaks.

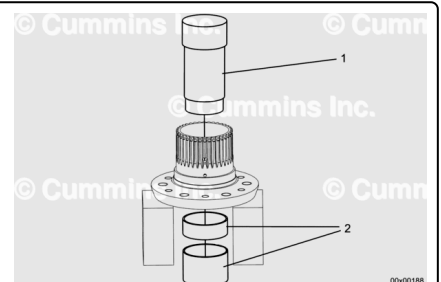
- Clamp the fan mounting hub in a bench vice. Clamp on the fan mounting flange.
- Indent the wear sleeve with a chisel in three equally spaced locations around the sleeve to loosen the press fit.
- Remove the wear sleeve.



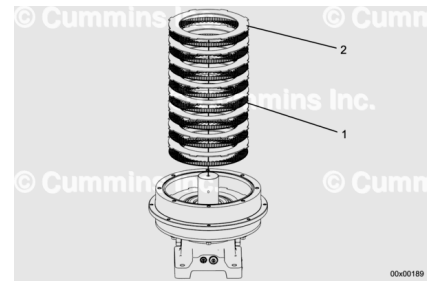
- Examine the two fan hub sleeve bearings for wear, indentations, nicks, or burs.
- Sleeve bearings in good condition can be re-used.
- Sleeve bearings in poor condition **must** be replaced.



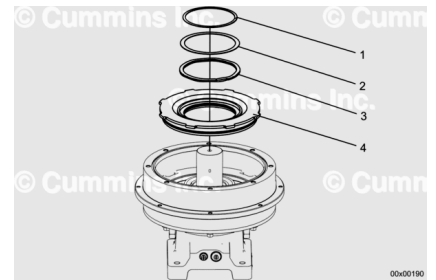
- If required, support the fan mounting hub under the flange.
- Use a hydraulic press and tool 1C (1) to remove fan mounting hub sleeve bearings (2).



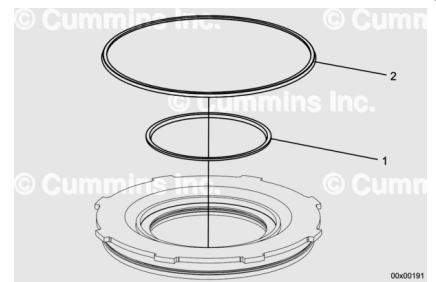
- Remove the facing plates (1) and steel clutch plates (2) from the main fan clutch assembly.
- Depending on Rockford® Clutch model, either a quantity of seven or eight of each plate will be removed.
- Record the quantity of removed plates to verify the same quantity are installed during rebuild.



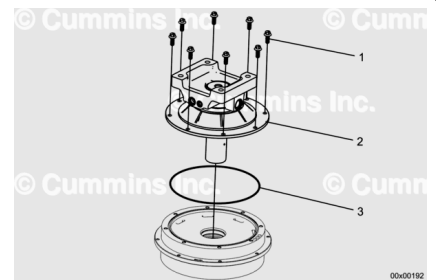
- Remove the external snapping (1), shim (2) and spring washer (3).
- Use two picks to pry the piston (4) upwards. Use gentle and consistent pressure to overcome residual oil suction from behind the seal rings.
- Remove the piston.



- Remove the inner (1) and outer (2) "L-shaped" seal rings from the piston.
- Use a seal pick. Discard the sealrings.

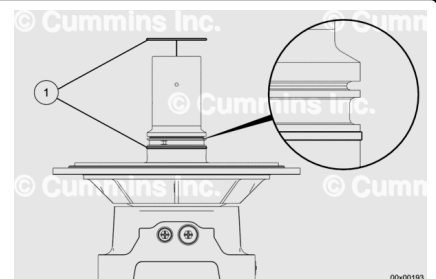


- Rotate the assembly and position face down on the working surface.
- Remove the pulley/adaptor capscrews (1). Lift the shaft and rear bearing retainer assembly (2) from the pulley/adaptor.
- Remove and discard the O-ring seal (3).



Note : Some brackets have metric threads. Do not damage the threads by using the incorrect eyebolts.

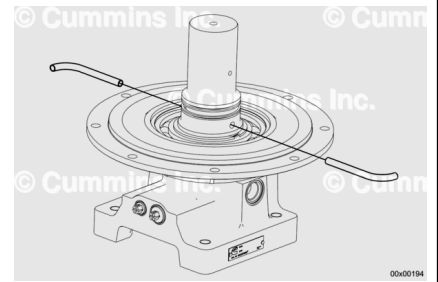
- Remove hook-type sealrings (1).
- Use a pick to un-hook the clasp on each sealring.
- Expand and remove from the grooves.
- Discard the sealrings.



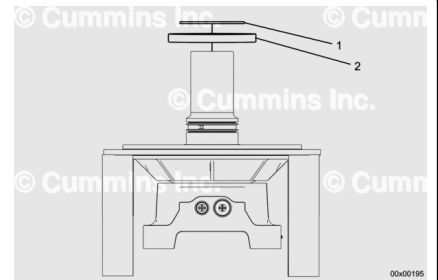
- Remove the pitot tubes. Use a screwdriver inserted into the end of the pitot tube to break it free of the Loctite®. Use

pliers to grip and remove.

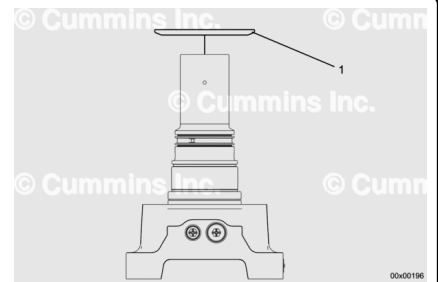
- Retain the pitot tubes for a length reference during assembly.
- Do **not** install used pitot tubes during rebuild. Discard used pitot tubes once a length reference is obtained.



- Remove the shaft assembly from the rear main bearing retainer.
- Support the rear main bearing retainer near the centre. Remove the external snapping (1).
- Use a hydraulic press and tool 3 (2) to press the shaft assembly out of the rear main bearing retainer.



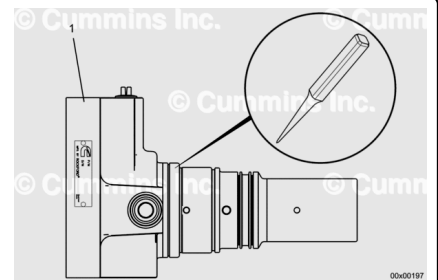
Using a prybar, remove the seal retainer (1) from the shaft assembly.



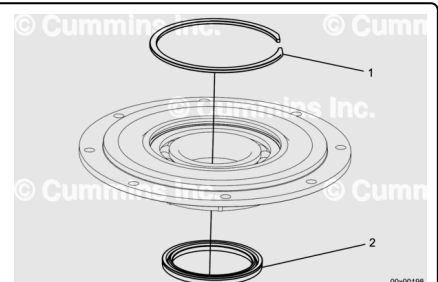
⚠ CAUTION ⚠

Do not cut through the wear sleeve with the chisel. Damage to the shaft will occur, causing seal leaks.

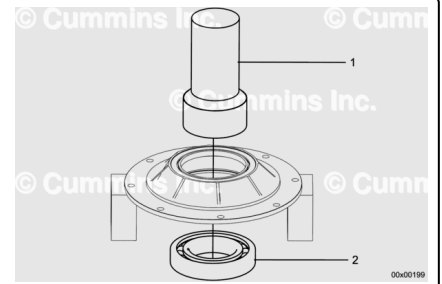
- Remove the shaft wear sleeve.
- Clamp the shaft assembly in a bench vice. Clamp on the mounting hub (1).
- Indent the wear sleeve with a chisel in three equally spaced locations around the sleeve to loosen the press fit.
- Remove the wear sleeve.



Remove the internal snapping (1) and the oil seal (X) from the rear main bearing retainer.



- Remove the rear main bearing.
- Support the rear main bearing retainer.
- Use a hydraulic press and tool 2 (1) to press the rear main bearing (2) out of the rear main bearing retainer.



Clean and Inspect for Re-use

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

- Use a scraper, wire brush, or Scotch-Brite™ pad to clean the machined gasket faces. Remove all sealant from pitot tube bores in the main shaft.
- Clean all parts using a steam cleaner and/or solvent. Cummins Inc. does **not** recommend any specific cleaning solution.
- Dry the components with compressed air.

- Visually inspect components for damage/wear. Check for cracks/burs/nicks on machined faces and seal grooves.
- Verify snap ring edges are square at the outer diameter.
- Confirm the snap rings are flat.

The following components may be re-used based on successful inspection:

- Capscrews
- Washers
- Retainer/Seal Assemblies
- Shaft
- Pulley/main body

- Snap rings (internal and external)

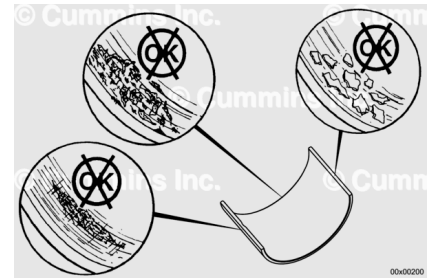
The following components **must** be replaced at rebuild:

- Seal rings
- Oil Seals
- Wear Sleeves
- Front and Rear Main Bearings

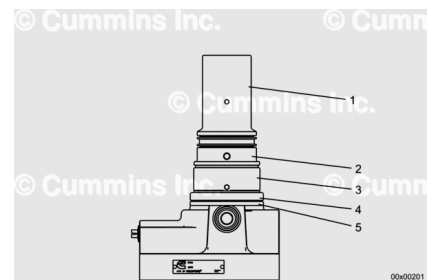
Inspect the sleeve bearings.

Replace any bearings that have the following damage:

- Nicks
- Cracks
- Burs
- Scratches (deep enough to be felt with a fingernail)
- Scrapes
- Gouges
- Embedded particles
- Fretting
- Pitting
- Flaking
- Corrosion
- Discolouration

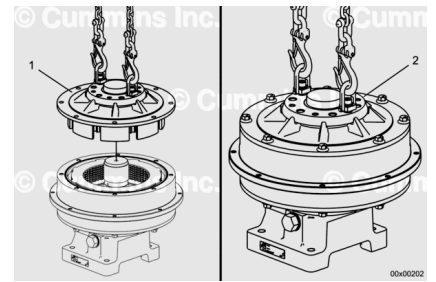


- Inspect the shaft and bracket assembly. Some shaft and bracket assemblies are constructed for manufacturing purposes as two piece units. Do **not** separate the shaft from the bracket.
- Verify that the shaft surface is smooth, with no nicks, burs or steps.
- Measure the diameter at specified points along the stepped shaft. Measure at least 4 points around the circumference. Measure multiple locations axially on shaft sections.
- Verify the shaft diameters:
 1. 69.799 mm [2.7480 in]
 2. 84.684 mm [3.3340 in]
 3. 95.032 mm [3.7414 in]
 4. 101.562 mm [3.9985 in]
 5. 104.912 mm [4.1304 in]

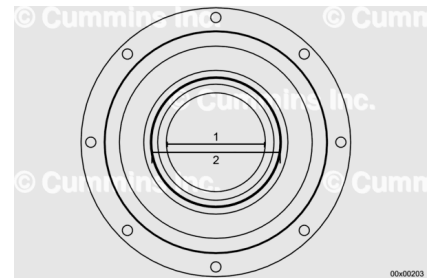


Note : Some Rockford® Clutch models have an engine lubricating oil hole with a roll pin in the hole. Verify that the engine oil hole is **not** plugged with debris.

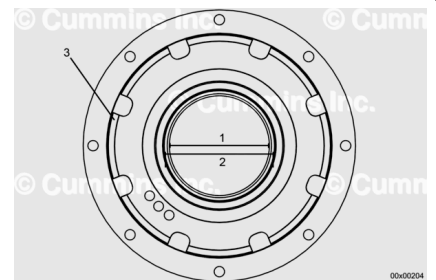
- Inspect the pulley/adapter; single piece (1) or two piece (2).
- Verify there are no cracks, nicks, burs or thread damage.



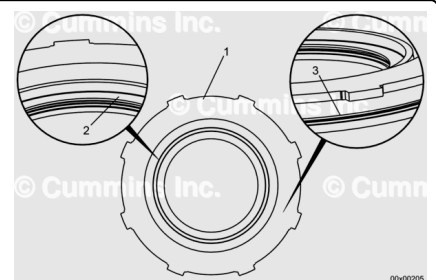
- Inspect the rear bearing retainer.
- Check that there are no nicks or burs rising above the mating face sealing surfaces or bearing interference fit surfaces.
- Check that the main bearing counterbore has a square bottom and is **not** oval from wear.
- Measure the rear main bearing bore.
 1. 130.213 mm [5.1265 in] MAX
 2. 169.979 mm [6.6921 in] MAX



- Inspect the front bearing retainer.
- Check that there are no nicks or burs rising above mating faces or sealing surfaces (1), bearing interference fit surfaces (2) or clutch engagement tangs (3).
- Check that the main bearing counterbore has a square bottom and is **not** oval from wear.
- Measure the front main bearing bore.
 1. 130.213 mm [5.1265 in] MAX
 2. 144.981 mm [5.7079 in] MAX



- Inspect the piston.
- Check that there are no nicks or burs rising above mating faces, sealing surfaces or clutch engagement tangs (1).
- Check the clutch plate contact face for flatness. The surface **must** be flat within 0.076 mm [0.003 in].
- Check the seal ring grooves (2 and 3) for any nicks, burs or debris which could affect the seal rings seating.
- Measure the piston thickness between the top and bottom machined faces.

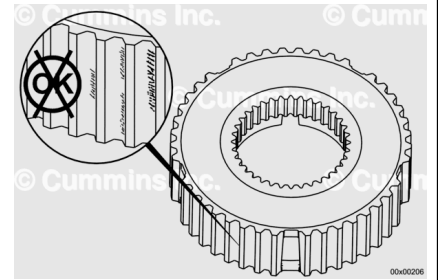


29 mm [1.141 in] MIN

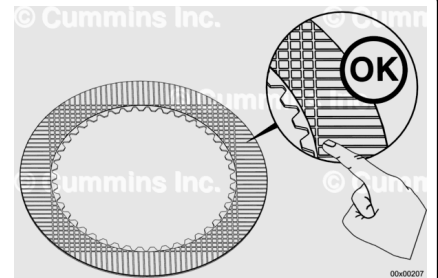
31 mm [1.221 in] MAX

- Inspect the clutch hub.
- Check that there are no grooves, nicks, or burs.

- Check the outer teeth for wear marks/grooves (1). Grooves will restrict clutch plate movement and disable the clutch from engaging.
- The clutch hub **must** be replaced if excessive wear or grooves are found on the outer teeth.



- Inspect the facing plates and steel clutch plates.
- Facing plates **must** be replaced when they are worn level with the grooves in the clutch material. Verify teeth are **not** excessively worn.
- Steel clutch plates **must** be of a minimum thickness and within flatness specifications.
- Flatness within 0.127 mm [0.005 in]
- Minimum thickness: 3.0734 mm [0.121 in]



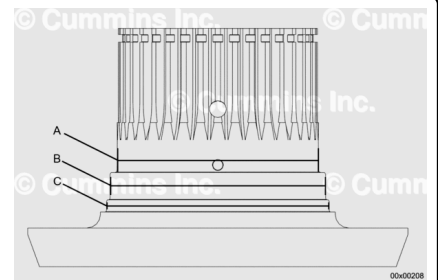
Note : If a service parts kit containing new facing and clutch plates has been ordered, this step can be omitted.

- Inspect the fan mounting hub.
- Check that there are no nicks or burs rising above sealing surfaces.

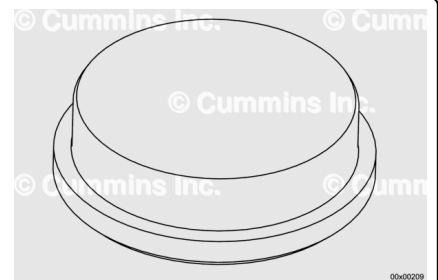
A. 95.011 mm [3.7406 in] MIN

B. 101.562 mm [3.9985 in] MIN

C. 104.912 mm [4.1304 in] MIN



- Inspect the end cap.
- Check for unusual wear, grooves, nicks or burs.



Assemble

- Assembly **must** be performed in a clean environment using the appropriate tooling and equipment designed for the size and weight of the components.
- Failure to use the specified tooling can result in oil leaks, bearing malfunction or other malfunction of the Rockford®

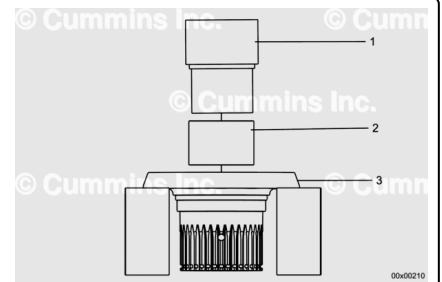
Clutch assembly.

©Cummins Inc.

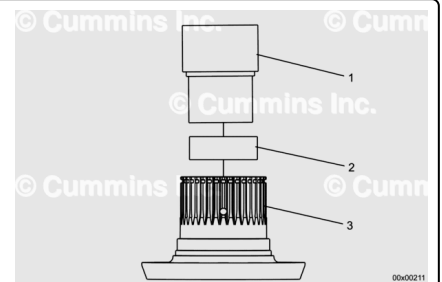
- Freeze the end cap.
- Use dry ice or place in a freezer for 30 minutes.

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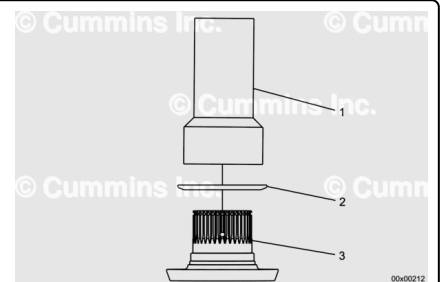
- If removed, install the front sleeve bearing (2) into the fan mounting hub (3).
- Use a hydraulic press and tool 1A (1) to press the sleeve bearing into the fan mounting hub bore.
- The orientation of the slit in the sleeve bearing is **not** significant and can be installed in any orientation.



- Install the rear sleeve bearing (2) into the fan mounting hub (3).
- Use a hydraulic press and tool 1B (1) to press the sleeve bearing into the fan mounting hub bore.
- The orientation of the slit in the sleeve bearing is **not** significant and can be installed in any orientation.

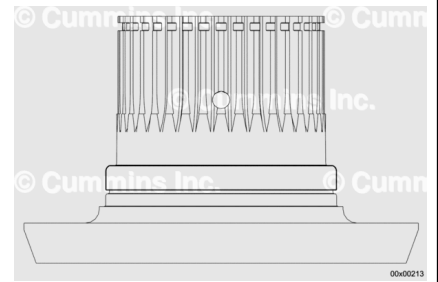


- Install the front seal retainer onto the fan mounting hub.
- Use a hydraulic press and tool 2 (1) to press the front seal retainer (2) onto the fan hub shaft (3).
- Tool 2 is machined to ensure the seal retainer is installed to the proper depth.

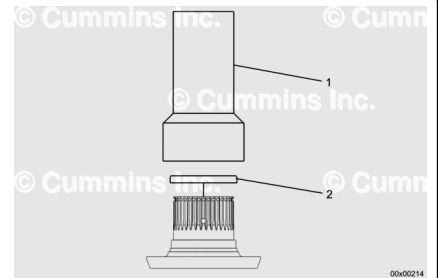


- The two wear sleeves are directionally machined and **must** be fitted in the correct position. Fitting the sleeves incorrectly **will** result in seal leaks.

- Wear sleeves colored blue **must** be fitted to the shaft and bracket assembly.
- Wear sleeves colored red **must** be installed on the fan mounting hub.

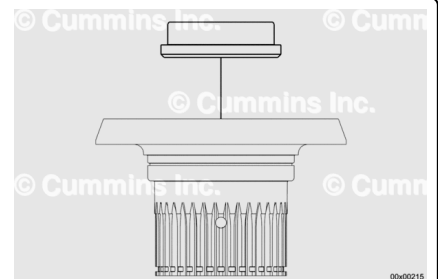


- Install the front wear sleeve (Red) onto the fan mounting hub.
- Apply Loctite® Primer N and Loctite® 242 or equivalent to the inner bore of the wear sleeve and the outer diameter of the fan mounting hub stub shaft.
- Use a hydraulic press and Tool 2 until the wear sleeve stops against the shoulder on the fan mounting hub.
- Do **not** polish, scratch or wear the surface of the wear sleeve. This will result in seal leaks.
- Wipe off excess sealant with a soft cloth.

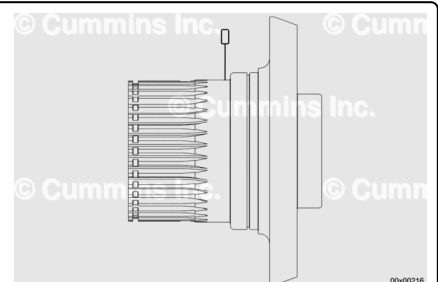


Note : Early fan mounting hubs were manufactured with a small lubricating oil hole under the wear sleeve. The new wear sleeve will cover the hole. The hole is **not** needed.

- If an end cap is used in this assembly, install the frozen end cap.
- Apply Loctite® Primer N and Loctite® 242 or equivalent to the outer diameter of the end cap and the inner bore of the fan mounting hub.
- Use the ram of a hydraulic press to install the end cap.
- Wipe off excess sealant with a soft cloth.

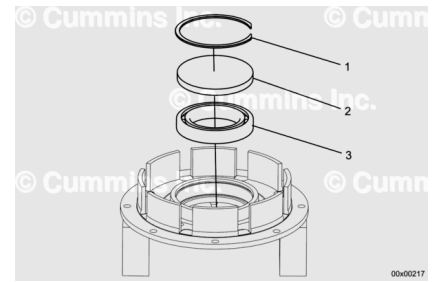


- Some clutch assemblies use a "pinned bearing" design.
- If the dowel pin was removed during cleaning, install into the fan mounting hub.
- Use the ram of a hydraulic press to install the dowel pin to a protrusion of 2.286 mm [0.090 in].

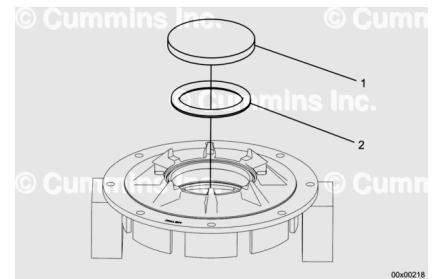


- Install the front bearing and snap ring.
- Use a hydraulic press. Place the front bearing retainer face down and support as close to the center as possible.

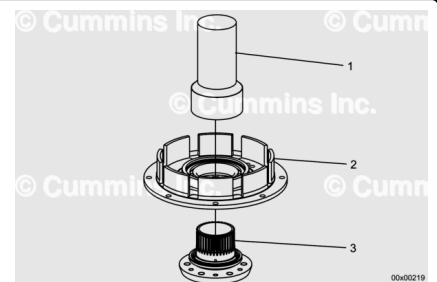
- Apply Loctite® Primer N and Loctite® 609 or equivalent to the outer bearing race and the inner bore of the front bearing retainer. Press the bearing into the bore using Tool 3.
- Pinned bearing designs **must** orient the bearing with the inner race pin notch facing downwards, entering the bore first.



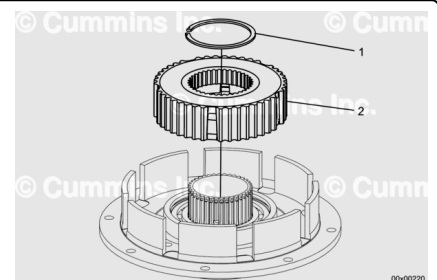
- Install the front seal.
- Use a hydraulic press. Place the front bearing retainer face up and support as close to the center as possible.
- Apply Loctite® Primer N and Loctite® 242 or equivalent to the outer seal diameter and the inner bore of the front bearing retainer. Verify sealant does **not** contact the seal inner diameter lip.
- Press the seal into the bore using tool 3. Wipe off excess sealant and rubber debris after installation.



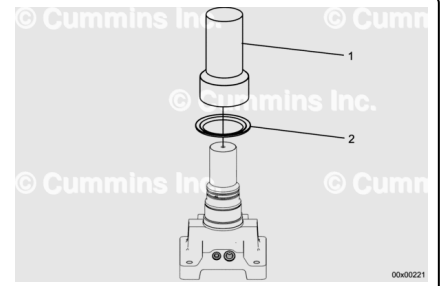
- Install the fan mounting hub assembly onto the front bearing retainer assembly.
- Apply Loctite® Primer N and Loctite® 609 or equivalent to the inner race of the front bearing in the front bearing retainer.
- Use a hydraulic press. Use tool 2 (1) to press the front bearing retainer (2) onto the fan mounting hub (3). Press until the bearing contacts the wear sleeve on the fan mounting hub.
- For pinned bearing designs, the notched inner bearing race **must** be aligned with the dowel pin in the fan mounting hub.
- Verify sealant does **not** contact the seal inner diameter lip.
- Once pressed together, spin the bearing retainer (approximately 25 revolutions) to burnish the seal lip to wear sleeve contact area.



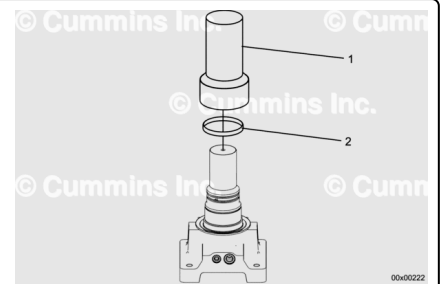
- Install the clutch hub.
- With the front retainer assembly face down, lower the clutch hub into position open end facing down.
- There is no specific orientation requirement for the installation of the clutch hub. Lubricating oil holes in the splines of the fan mounting hub are **not** relevant.
- Install the external snap ring (1) to retain the clutch hub (2).



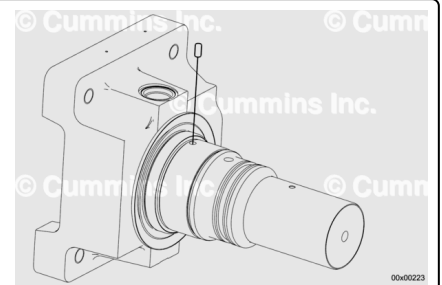
- Install the rear seal retainer onto the shaft and bracket assembly.
- Use a hydraulic press. Position the shaft and bracket assembly with the shaft facing upwards. Use tool 2 (1) to press the rear seal retainer (2) onto the shaft.
- Tool 2 is machined to ensure the seal retainer is installed to the proper depth.



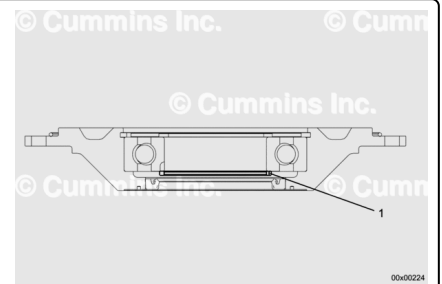
- Install the rear wear sleeve (blue) onto the shaft and bracket assembly.
- Apply Loctite® Primer N and Loctite® 242 or equivalent to the inner bore of the wear sleeve and the outer diameter of the fan mounting hub stub shaft.
- Use a hydraulic press and tool 2 (1) until the wear sleeve (2) stops against the shoulder on the fan mounting hub.
- Do **not** polish, scratch or wear the surface of the wear sleeve. This will result in seal leaks.
- Wipe off excess sealant with a soft cloth.



- Some clutch assemblies use a pinned bearing design.
- If the dowel pin was removed during cleaning, install the dowel pin into the fan mounting hub.
- Use the ram of a hydraulic press to install the dowel pin to a protrusion of 2.286 mm [0.090 in].



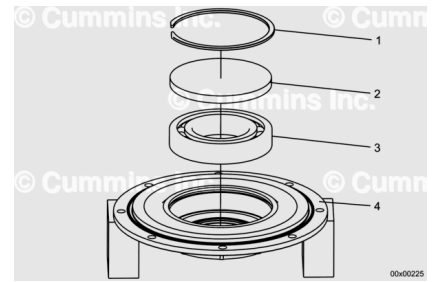
- Clutches with large rear main bearing configurations may use a bearing spacer ring. If a spacer ring was removed during disassembly, install in the same position.
- If a clutch uses a “pinned bearing design”, a notched spacer (1) is used. The notched spacer **must** be placed in the rear bearing retainer before pressing in the rear main bearing.
- The notched spacer **must** be aligned to the dowel in the shaft assembly later in this procedure.



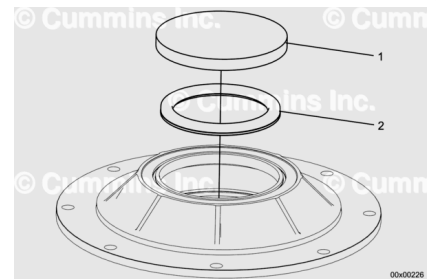
- Install the rear bearing (3) into the rear bearing retainer (4).
- Use a hydraulic press. Place the rear bearing retainer rear face up and support as close to the center as possible.
- Apply Loctite® Primer N and Loctite® 609 or equivalent to the outer bearing race and the inner bore of the rear

bearing retainer.

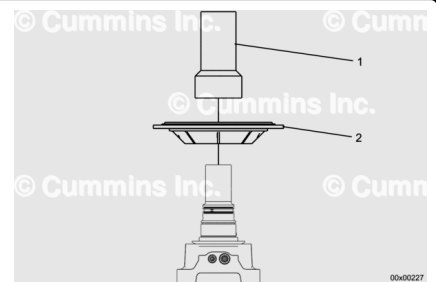
- Press the bearing into the bore. For small rear bearings, use Tool 3. For large rear bearings, use tool 4 (2).
- Pinned bearing designs **must** orient the bearing with the inner race pin notch facing downwards, entering the bore first.
- Install the internal snap ring (1).



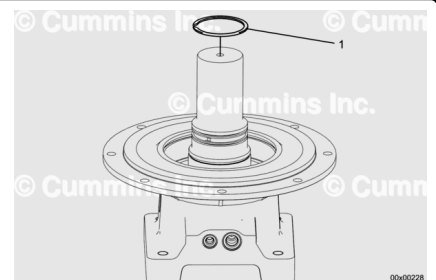
- Install the rear oil seal (2).
- Use a hydraulic press. Place the rear bearing retainer rear face down and support as close to the center as possible.
- Apply Loctite® Primer N and Loctite® 242 or equivalent to the outer seal diameter and the inner bore of the front bearing retainer. **Never** allow sealant to contact the seal inner diameter lip.
- Press the seal into the bore using tool 3 (1). Wipe off excess sealant and rubber debris after installation.



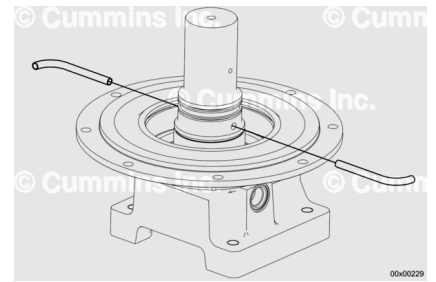
- Install the rear bearing retainer assembly on to the shaft and bracket assembly.
- Apply Loctite® Primer N and Loctite® 609 or equivalent to the inner race of the rear bearing in the rear bearing retainer.
- Use a hydraulic press. Use tool 2 to press the rear bearing retainer onto the shaft and bracket assembly. Press until the bearing contacts the wear sleeve.
- Pinned bearing designs, the notched inner bearing race and notch in the spacer ring **must** be aligned with the dowel pin in the shaft and bracket assembly.
- Verify sealant does **not** contact the seal inner diameter lip.
- Once pressed together, spin the bearing retainer (approximately 25 revolutions) to burnish the seal lip to wear sleeve contact area.



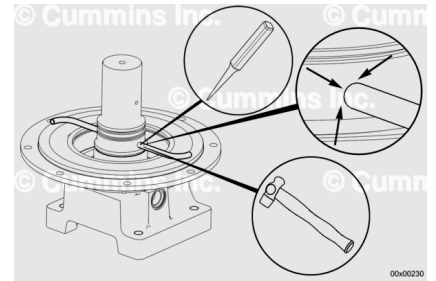
- Install the external snap ring.
- Verify that the snap ring is properly seated in the groove. Tapping with a small hammer may be required to pop snap ring into the groove.



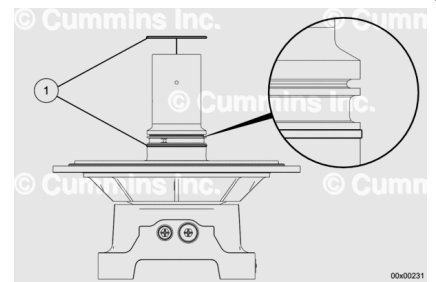
- Install the pitot tubes. Cut to length if necessary. Use the old tubes as a length reference.
- Apply Loctite® Primer N and Loctite® 609 or equivalent to outer diameter of the tubes.
- Push pitot tube to the bottom of the bore in the shaft and bracket assembly. Orient each tube in the direction shown, parallel to the surface of the bearing retainer, opposing each other. A screwdriver inserted into the pitot tube can be used to align.



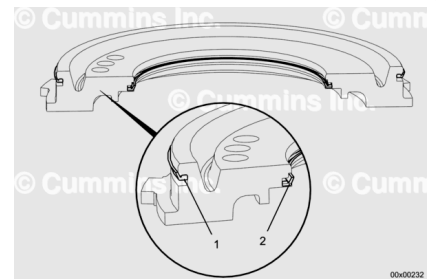
Stake the tubes in position. Orient each tube correctly and stake at three places (9, 12, and 3 o'clock positions) to prevent rotation during operation.



- Install the two hook-type seal rings (1).
- Once installed, rotate the seal rings such that their joins are opposing each other (180 degrees apart).



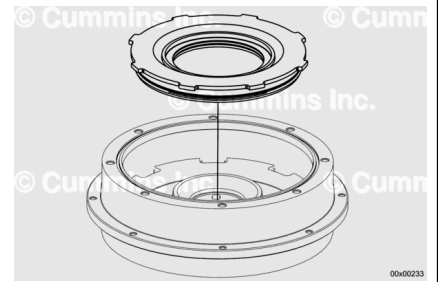
- Assemble the lip-type seal rings into the two grooves on the piston.
- Lubricate the seal ring grooves with an engine grease or equivalent oil-soluble lubricant.
- Install the outer (larger) seal ring (1) in the outer groove of the piston.
- Install the inner (smaller) seal ring (2) in the inner groove of the piston.
- Both seal rings **must** have the thick square section of the seal filling the groove of the piston, with the "lip" protruding to seal against mating components.



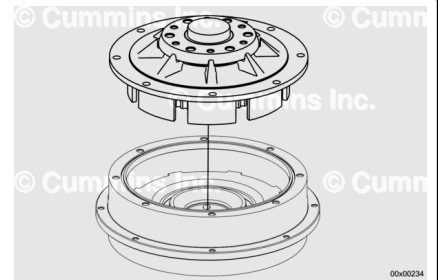
- Install the piston in the pulley/adapter body.
- Lubricate the internal surfaces of the pulley/adapter body with an engine grease or equivalent oil-soluble lubricant.
- Gently place the piston in the pulley/adapter body. The piston **must not** be forced into position. Excessive force

will roll or cut the lip type seal rings and cause oil leaks during operation.

- Without pressing down on the piston, rotate the piston slowly alternating clockwise to counterclockwise until it falls into place.

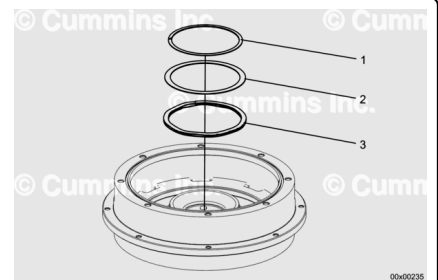


- To facilitate final assembly, pre-align the piston in the pulley/adapter body.
- Place the front bearing retainer assembly in place in the pulley/adapter body. Align the tangs of the piston with the slots of the bearing retainer so the bearing retainer sits flush in its “assembled” position on the pulley adapter/body.
- Rotate the bearing retainer (and thus the piston) until the outer ring of bolt holes on the bearing retainer are aligned with the pulley/adapter body. This positions the piston so the tangs will be aligned during final assembly.
- Gently remove the front bearing retainer to retain the alignment of the piston.

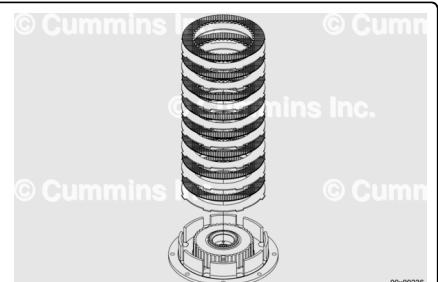


Note : Determine the proper thread size when using eyebolts. Some clutches have metric threads.

- Install the spring washer (3), shim (2) and external “spirolock” snap ring (1) to retain the piston.
- Push down to compress the spring washer while feeding the “spirolock” snap ring into the groove.
- If there is excessive force required to compress the spring washer and shim, the seals on the piston may have turned/cut and are disturbing the piston from its normal seated position in the pulley adapter.
- Remove the piston using lifting hooks and check the seals. Re-install if seal condition is deemed serviceable.



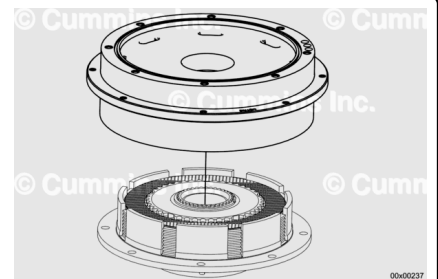
- Install the facing plates and steel clutch plates in position in the front bearing retainer assembly. The front bearing retainer assembly **must** be placed face down and supported for balance.
- Submerge the facing plates in new, clean engine oil and allow the excess oil to drain off before installing.
- Install one steel clutch plate in place in the front bearing retainer. Then place one facing plate (FD 30) on top of the steel clutch plate. Alternately stack the remaining plates



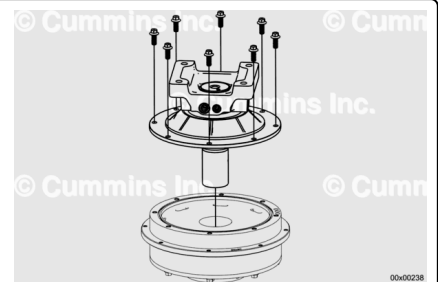
until a total of 7 or 8 of each have been placed on the stack, and the top plate is a facing plate.

- Some fan clutches use 7 of each plate and some use 8. If 8 of each plate are installed on a clutch that requires 7 of each plate, the pulley/pulley adapter will **not** sit properly and will **not** bolt up to the bearing retainer assembly.

- Rotate the pulley/adapter body so it can be lifted to face vertically downwards. Use appropriate eyebolts and lifting equipment.
- Install a guide bolt in one hole of the pulley/adapter body.
- Stick the pulley/adapter body "O"-ring into the groove using grease.
- Gently lower the pulley/adapter body onto the bearing retainer assembly. Align the guide bolt with one of the holes of the front bearing retainer assembly. The piston tangs will align with the slots in the bearing retainer as the pulley/adapter mating face contacts the bearing retainer.
- Remove eyebolts. Tighten four capscrews hand tight.



- Install and tighten the pulley/adapter body capscrews with washers, pulling the mating faces together with 4 opposite bolts first.
- Remove the guide bolt.
- Tighten all capscrews.



If capscrews are 3/8"-16 UNF:

Torque Value: 54 n•m [40 ft-lb]

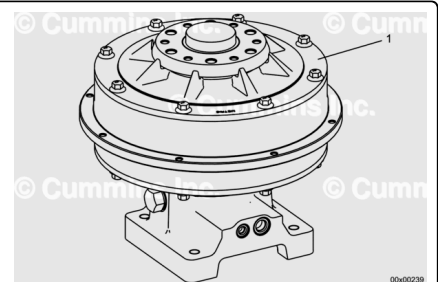
If capscrews are M10 x 1.50:

Torque Value: 61 n•m [45 ft-lb]

If capscrews are M12 x 1.75:

Torque Value: 102 n•m [75 ft-lb]

- Rotate the assembly on a bench, such that the fan mounting hub is upwards.
- Install remaining capscrews and washers (1).
- Tighten all capscrews.



If capscrews are 3/8"-16 UNF:

Torque Value: 54 n•m [40 ft-lb]

If capscrews are M10 x 1.50:

Torque Value: 61 n•m [45 ft-lb]

If capscrews are M12 x 1.75:

Torque Value: 102 n•m [75 ft-lb]

- If the pulley is a 2-piece bolt-on design, install the pulley.
- Install the outer ring of bolts and washers.
- "Heavy side" indicators may be stamped on the flanges of the pulley and/or pulley adapter. They are shown with the letter "H". These **must** be located 180° apart to obtain the best balance.
- Tighten all capscrews.

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If capscrews are 3/8"-16 UNF:

Torque Value: 54 n•m [40 ft-lb]

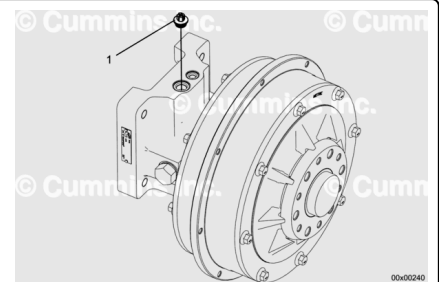
If capscrews are M10 x 1.50:

Torque Value: 61 n•m [45 ft-lb]

If capscrews are M12 x 1.75:

Torque Value: 102 n•m [75 ft-lb]

- If the oil feed port orifice fitting was removed during disassembly, re-install.
- Location of the oil feed port varies between clutch models. Verify the port is "In" port before installing orifice fitting.



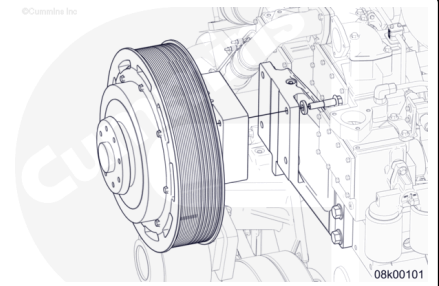
Testing

- Rockford® Clutch post-rebuild testing is to check for oil leaks **only**.
- If the rebuild has taken place at a customer or field servicing location, it is possible to test the clutch on engine.
- If the rebuild has taken place at an official rebuild center, it is possible to test the clutch on a dedicated test rig. The rig is **not** purchasable and **must** be built specifically to perform Rockford® Clutch testing. To understand the details of test rigs already in operation, contact your regional service engineering support.

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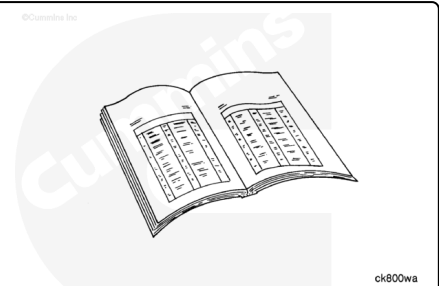
Install

- Install the Rockford ® Clutch. See corresponding Service Manual. Reference Procedure 008-036 in Section 8.
- Operate the clutch engaged and under load for one hour. Stop the engine and visually inspect for oil leaks.
- If no oil leaks are observed, clutch can be permanently installed.
- If oil leaks are observed, attempt to identify the source of the leak. The clutch **must** be removed and torn down to replace leaking seal or seal ring.

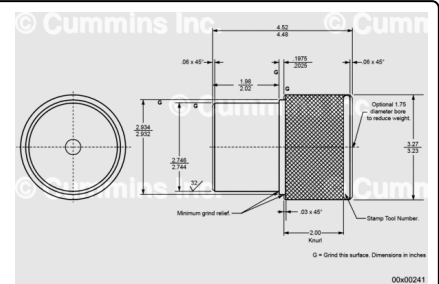


Overhaul Tools

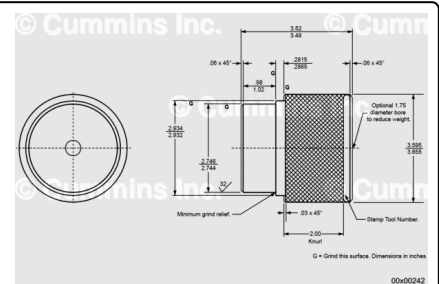
Rockford Clutch® Service Tools **must** be made by a competent engineering shop to the specification outlined in the technical drawings.



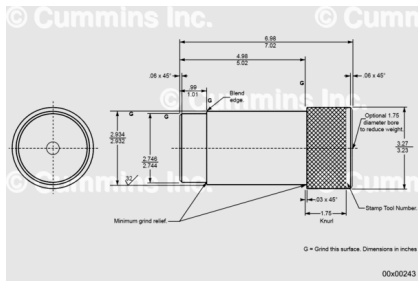
Tool 1A: Front sleeve bearing installer.



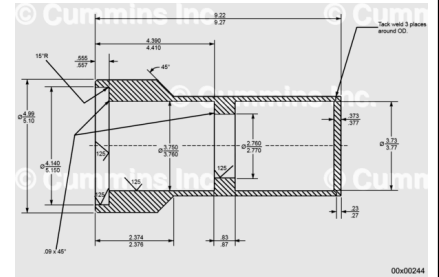
Tool 1B: Rear sleeve bearing installer.



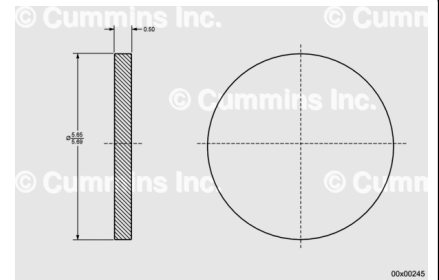
Tool 1C: Front and rear sleeve bearing remover.



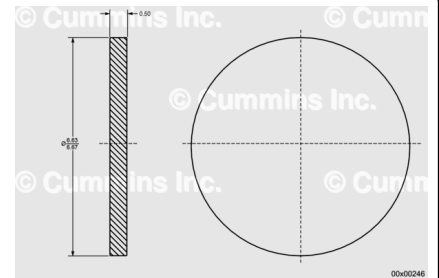
Tool 2: Wear sleeve and retainer/seal assembly installer.
Bearing (small and large) remover. Assembly pusher tool.



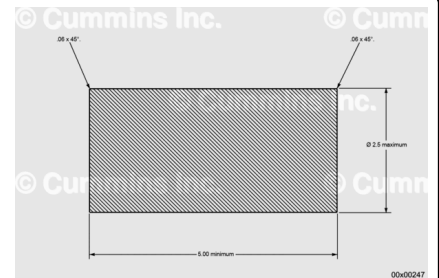
Tool 3: Bearing (small diameter) installer.



Tool 4: Bearing (large diameter) installer.



Tool 5: End cap remover.



Document History

Date	Details
2025-5-6	Module Created

Last Modified: 08-Jul-2025
